

ECON 403/503—INTRODUCTION TO CONTRACT THEORY AND AUCTIONS

Fall 2020

Department of Economics
Nazarbayev University

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Lecture Delivery Mode: Asynchronous delivery through recorded lectures plus additional group office hours, please see below.

Office hours: By appointment, book weekly at <https://alejandromelo.youcanbook.me> during available slots, please sign at least one day in advance.

Course Description This course analyzes problems and issues created by informational asymmetries and how properly designed contracts could ameliorate those problems. In other words, this course takes a first look at the theory of incentives, which is at the core of modern microeconomics and game theory. Topics include adverse selection, screening, signaling, and moral hazard, as well as various applications to insurance, labor and credit markets, auctions, and organizational design. I will also introduce some formal instruments: how to work with uncertainty, equilibrium concepts in static and dynamic games and games with incomplete information. If time permits, we will look also at the basic theory of mechanism design.

Course Requirements ECON 201 Intermediate Microeconomics (C- and above) and ECON 301 Econometrics I (C- and above). The theory of incentives analyzes the design of optimal contracts and institutions in environments with asymmetric information and conflict of interest. The main mathematical tool used in the analysis is nonlinear optimization under constraints. We will briefly review the methods of constrained optimization if needed, but I will assume that you have a good background in calculus and probability. It is crucial that you are comfortable with analytical reasoning and mathematical notation in order to be able to follow the lectures and the readings.

Course Material There is no textbook required for this course. The course is based mainly on my lectures, slides and readings. All materials will be uploaded to Moodle.

The following is a list of undergraduate and graduate textbooks that you may find useful:

- Bolton and Dewatripont (2005), Contract Theory, MIT Press.
- Laffont and Martimort (2002), The Theory of Incentives, PUP.
- Salanie (1997), The Economics of Contracts: A Primer, MIT Press.
- Milgrom (2003), Putting Auction Theory to Work, Cambridge University Press.
- Krishna (2009), Auction Theory, Academic Press.

Grading Policy Grading will be weighted as follows:

For ECO403 (UG students)	For ECO503 (MA students)
Essay (15%)	Referee Report (15%)
Problem Sets (20%)	Problem Sets (20%)
Midterm (30%)	Midterm (30%)
Final exam (35%)	Final exam (35%)

Class delivery The mode of the course will be **asynchronous delivery**. This means that we will not hold virtual real time lectures, but instead all of the content will be delivered through prerecorded lectures and reading materials, which will be uploaded to Moodle each week. The only exception to this rule may be our first lecture, in which I will introduce myself and go over some of the specifics of the logistics the course. However, we will hold regular **group office hours**, at a minimum of once every two weeks, so that I can meet with you and go over some questions that you may have about the course material. These group office hours will be scheduled at a time that is convenient for all students; these sessions will not necessarily be scheduled at the official time of the class as posted in the Office of the Registrar.

Problem Sets We will have 5 problem sets. All problem sets will be submitted through the Moodle platform. You will be required to submit your write up as a **single pdf file**. You will either need to scan your solutions and combine the scans into a single pdf or you will need to type your solutions.

Essay Undergraduate students have to write one essay where they describe some situation that happened to them, or that happened recently in the world or in some TV-series (or other piece of popular culture) that can be analyzed using one of the models we have studied. You have to write down a very simple model of this situation and then solve it. Finally, you need to comment about the results you got. Does the model predicts what happened in reality? Tentative deadline: December 6

Midterm The tentative date will be on the week of October 14, depending on the progress during the course . It will cover roughly half of the material in the course. The midterm will be a take-home exam which you will have 24 hours to solve it. Your submission will consist of *two parts*: a pdf of your solutions and a short video, no longer than 10 minutes, in which you will record yourself explaining your solutions to me.

Referee Report Master students have to prepare a referee report about a paper that I will assign. I will assign papers after the midterm and give you more details about the assignment in due time.

Final Exam Will cover the second half of the material. The date will be chosen by the Registrar, during Finals Week. The style of the exam is similar to the midterm, a take-home exam which you will have 24 hours to solve. Once again, the submission will consists of a pdf of your solution and a short video in which you will explain your answers to me.

A letter grade based on the sum of the components above will be assigned as follows:

Letter Grade	Score Achieved
A	[95, 100]
A–	[90, 95)
B+	[85, 90)
B	[80, 85)
B–	[75, 80)
C+	[70, 75)
C	[65, 70)
C–	[60, 65)
D+	[55, 60)
D	[50, 55)
F	[0, 50)

Course attendance As mentioned earlier, the main mode of delivery is asynchronous, so I will not take course attendance. I will try to make the course as easy as possible to follow by organizing it through the Moodle platform. However, you will have to work extra hard to keep on track with the course material. The group office hours will also serve for me to get an idea of how you are doing. It will be **IMPERATIVE** that you strive to prepare yourself before these office hours, so that you come armed with as many

clear questions as possible. If you need more individualized help, I will be happy to meet you during individual office hours, which can be scheduled through <https://alejandromelo.youcanbook.me> during available slots. Please book at least 24 hours in advance.

Make-up Policy For each regular exam, a make-up exam will be granted only once and only in case of properly documented emergencies, such as a medical report from the school medical center. Students who want to take a make-up exam need to inform me by email as early as possible but no later than the day after the regular exam. Make-up for the midterm exam will be given within a week after the exam date. Make-up for the final exam will be offered during after finals week and in accordance with the Registrar's policy. There will be no make-up for the make-up.

Rules and Regulations During lectures, please keep your laptops, tablets and cell phones switched off and refrain from text messaging. Do not arrive late or leave early without prior permission. Do not engage in side conversations or other disruptive behaviors.

Academic Misconduct Academic misconduct is an act in which a student:

1. seeks to claim credit for the work or efforts of another without authorization or citation ("plagiarism");
2. uses unauthorized materials or fabricated data in any academic exercise;
3. forges or falsifies academic documents or records;
4. intentionally impedes or damages the academic work of others;
5. engages in conduct aimed at making false representation of any student's academic performance;
6. engages in any form of cheating on an exam or assignment;
7. assists other students in any of these acts

These are serious offenses subject to disciplinary action depending on their severity as detailed in the Student Code of Conduct.

Tentative Course Outline

- Uncertainty. Expected utility, risky assets, risk aversion.
- Math review: constrained optimization.
- Asymmetric information: adverse selection, screening, competitive screening.
- Asymmetric information: signaling.
- Asymmetric information: moral hazard.
- Basic principal-agent model.
- Auctions, optimal auctions.
- Introduction to mechanism design.

Disclaimer Subsequent changes can be made to any part of this syllabus when necessary. Any changes will be announced in class and on Moodle.